UNIVERSIDAD NACIONAL DE INGENIERÍA

Facultad de Ciencias

Escuela Profesional de Ciencia de la Computación

EXAMEN FINAL

Curso: **Fundamentos de Programación CC112 - A**

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1.

```
test01@pcs5:~/Documentos/PC02/Semana08/FINAL$ valgrind ./build/debug/problema1
==6753== Memcheck, a memory error detector
==6753== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==6753== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==6753== Command: ./build/debug/problema1
==6753==

Ingrese el tamaño de las filas: 3
Ingrese el tamaño de las columnas: 4

Matriz M:
1    2    3    4
2    4    6    8
3    6    9    12

Vector resultante: { 10 20 30 };

==6753==
==6753== HEAP SUMMARY:
==6753==     in use at exit: 0 bytes in 0 blocks
==6753==   total heap usage: 8 allocs, 8 frees, 75,860 bytes allocated
==6753==
==6753== All heap blocks were freed -- no leaks are possible
==6753==
==6753== For lists of detected and suppressed errors, rerun with: -s
==6753== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

2.

```
test01@pcs5:~/Documentos/PC02/Semana08/FINAL$ valgrind ./build/debug/problema2
==7117== Memcheck, a memory error detector
==7117== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==7117== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==7117== Command: ./build/debug/problema2
==7117==

Punto p1:
Ingrese la coordenada x: 3
Ingrese la coordenada y: 5

Punto p2:
Ingrese la coordenada x: -5
Ingrese la coordenada y: 3

Ambos puntos estan a la misma distancia del origen.
==7117==
==7117== HEAP SUMMARY:
==7117==     in use at exit: 0 bytes in 0 blocks
==7117==   total heap usage: 9 allocs, 9 frees, 76,036 bytes allocated
==7117==
==7117== All heap blocks were freed -- no leaks are possible
==7117==
==7117== For lists of detected and suppressed errors, rerun with: -s
==7117== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

```
test01@pcs5:~/Documentos/PC02/Semana08/FINAL$ valgrind ./build/debug/problema2
==7160== Memcheck, a memory error detector
==7160== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==7160== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==7160== Command: ./build/debug/problema2
==7160==

Punto p1:
Ingrese la coordenada x: 5
Ingrese la coordenada y: -3.4

Punto p2:
Ingrese la coordenada x: -4.9
Ingrese la coordenada y: 7.2

El punto mas lejano es:
Punto p2: (-4.9, 7.2)

==7160==
==7160== HEAP SUMMARY:
==7160==     in use at exit: 0 bytes in 0 blocks
==7160==   total heap usage: 9 allocs, 9 frees, 76,036 bytes allocated
==7160==
==7160== All heap blocks were freed -- no leaks are possible
==7160==
==7160== For lists of detected and suppressed errors, rerun with: -s
==7160== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```